Saydul Akbar Murad | *Graduate Assistant*

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Permanent: Dokkhin Khil Chadok, Koiar Bil, Chakaria, Cox's Bazar, Bangladesh.
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Education

2023 – 2026 **Ph.D. Candidate in Computer Science**
University of Southern Mississippi, Mississippi, USA.
Research Topic: *Efficient Real-Time EEG-to-Text Decoding Using LDEPTH-Based P2P Fog Computing Architecture for Scalable Brain-Computer Interfaces.* Code

2021 – 2023 **Master of Science in System Network and Security (By Research)**
Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia
Research Topic: *A New Priority Rule Cloud Scheduling Technique That Utilizes Gaps to Increase the Efficiency of Jobs Distribution.* Code
Milestone: *Best Publication Award.*

2016 – 2020 **Bachelor of Science in Information & Communication Engineering**
Noakhali Science and Technology University, Noakhali, Bangladesh.
Project: *An Automated System to Maintain Departmental Activities Over the Cloud: An Android Application Approach.* Code

Research Interests

• Cybersecurity • ML • DL • LLM • EEG Signal Decoding • P2P Communication • Fog Computing

Honors And Awards

2026 **Graduate Student Hall of Fame**
Awarded by the Graduate School, University of Southern Mississippi, Mississippi, United States.

*2023 – ** **Fully Funded Scholarship & GA Award**
Awarded by The University of Southern Mississippi, Mississippi, United States.

2025 **Best Paper Award**
Awarded by 40th International Conference on Computers and Their Applications.

2025 **NAIRR Pilot Program Travel Grant**
Awarded full funding for travel, lodging, and meals to participate in the National AI Research Resource (NAIRR) Pilot program.

2023 **Best Publication Award**
Awarded by Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia.

2023 **Best Researcher Award**
Awarded by International Research Awards on Computer Aided Design in Mechanical Eng.

2021 – 2023 **Fully Funded Scholarship**
Awarded by Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia.

2020 **Best Paper Award**
Awarded by The 8th International Conference on Computer and Communications Management.

2016 **3rd Place at NSTU Intra University Programming Contest**
Awarded by Noakhali Science and Technology University, Noakhali, Bangladesh.

Funds/Grants/Fellowships Including Pending

2.1 Research Grants

2026 – 2030 **Submitted**

Title: Privacy-Preserving Distributed Fog-Cloud Computing for Real-Time Neural Signal Processing and Brain-Computer Communication
Role: Key contributor to the writing of the grant proposal.
PI: Dr. Nick Rahimi
Funding Body: National Science Foundation (NSF) **Amount:** $731,344

2025

Title: SecureBrainNet: A Privacy-Preserving Distributed Framework for Brain-Computer Interfaces
Role: Key contributor to the writing of the grant proposal.
PI: Dr. Nick Rahimi
Funding Body: University of Southern Mississippi (USM) **Award:** $4500

2.2 Research Fellowship

2025 **Dissertation Completion Grant Award**, USM **Award:** $2000
Title: *"Efficient Real-Time EEG-to-Text Decoding Using LDEPTH-Based P2P Fog Computing Architecture for Scalable Brain-Computer Interfaces"*

2024 **Best Proposal Project**, USM **Award:** $500
Title: *"FMRG-Bio: Future Nanotherapy Manufacturing-transitioning from the lab to the pharma factor"*

2021 **Master Research Scheme (MRS)**, UMPSA **Award:** RM21,600
Funded by Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia

2.3 Travel Grants

2024 **Graduate School's Travel Award**, USM **Award:** $500
Support for conference registration, attending, or presenting papers

2023 **School of BEES & CAS** , USM **Award:** $500
Support for conference registration, attending, or presenting papers.

Publications

*Google Scholar statistics: Total of **1021** citations as of February 27, 2026, with h-index = 15, i10-index = 18*
*Researchgate statistics: Total of **893** citations as of February 27, 2026, with h-index = 14*

Selected Journal Articles

1. **Saydul Akbar Murad**, Ashim Dahal, Nick Rahimi "*EEG-to-Text Translation: A Model for Deciphering Human Brain Activity*", arXiv, 2025
2. **Saydul Akbar Murad**, Ashim Dahal, Nick Rahimi "*Multi-Lingual Cyber Threat Detection in Tweets/X Using ML, DL, and LLM: A Comparative Analysis*", IEEE Transactions on Computational Social Systems, 2025
3. Ashim Dahal, Ankit Ghimire, **Saydul Akbar Murad**, Nick Rahimi "*Redemption Score: An Evaluation Framework to Rank Image Captions While Redeeming Image Semantics and Language Pragmatics*", arXiv, 2025

4. Ashim Dahal, **Saydul Akbar Murad**, Nick Rahimi "*Efficiency Bottlenecks of Convolutional Kolmogorov-Arnold Networks: A Comprehensive Scrutiny with ImageNet, AlexNet, LeNet and Tabular Classification*", arXiv, 2025
5. Ashim Dahal, **Saydul Akbar Murad**, Nick Rahimi "*Heuristical Comparison of Vision Transformers Against Convolutional Neural Networks for Semantic Segmentation on Remote Sensing Imagery*", IEEE Sensors Journal, 2025
6. **Saydul Akbar Murad,** Nick Rahimi *"Unveiling Thoughts: A Review of Advancements in EEG Brain Signal Decoding into Text*", IEEE Transactions on Cognitive and Developmental Systems, 2024
7. **Saydul Akbar Murad,** Nick Rahimi *"Secure and Efficient Hierarchical P2P Fog Architecture: A Novel Approach for IoT*", IEEE Internet of Things Journal, 2024
8. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, MD Khairul Bashar Bhuiyan, Md Saib, Nick Rahimi *"SG-PBFS: Shortest Gap-Priority Based Fair Scheduling technique for job scheduling in cloud environment*", Future Generation Computer Systems, 2024
9. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, Md Murad Hossain Sarker, M Saef Ullah Miah, MD Khairul Bashar Bhuiyan, Nick Rahimi, Anupam Kumar Bairagi *"Priority based job scheduling technique that utilizes gaps to increase the efficiency of job distribution in cloud computing*", Sustainable Computing: Informatics and Systems, 2024
10. Nusrat Jahan Prottasha, **Saydul Akbar Murad,** Abu Jafar Md Muzahid and Md. Kowsher *"A training-based algorithm for machine learning: Impact Learning"*, Journal of Computational Science, 2023
11. Abu Jafar Md Muzahid, Syafiq Fauzi Bin Kamarulzaman, Dr Md Arafatur Rahman, **Saydul Akbar Murad** *"Multiple Vehicle Cooperation and Collision Avoidance in Automated Vehicles: Survey and an AI-Enabled Conceptual Framework"*, Scientific Reports, 2023
12. Israt Jahan, K. M. Aslam Uddin, **Saydul Akbar Murad,** M. Saef Ullah Miah, Tanvir Zaman Khan, Mehedi Masud, Sultan Aljahdali, Anupam Kumar Bairagi *"4D: a real time driver drowsiness detector using deep learning*". Electronics, 2023
13. **Saydul Akbar Murad,** Abu Jafar Md Muzahid, and Zafril Rizal Bin M Azmi *"A Review on Job Scheduling Technique in Cloud Computing and Priority Rule Based Intelligent Framework*", JKSU - Computer and Information Sciences, 2022
14. **Saydul Akbar Murad,** ABU JAFAR MD MUZAHID and Apurba Adhikary *"AI Powered Asthma Prediction Towards Treatment Formulation: An Android App Approach"*, Intelligent Automation & Soft Computing, 2022
15. Apurba Adhikary, Joy Bhuiya, **Saydul Akbar Murad,** Md. Bipul Hossain, K. M. Aslam Uddin,MD Estihad Faysal, Abidur Rahaman, Anupam Kumar Bairagi *"Performance Evaluation of Micro Lens Arrays: Improvement of Light Intensity and Efficiency of White Organic Light-Emitting Diodes"*., PLOS ONE, 2022
16. Nusrat Jahan Prottasha, Abdullah Sami, MD Kowsher, **Saydul Akbar Murad,** Anupam Kumar Bairagi, Mehedi Masud, and Mohammed Baz *"Transfer Learning for Sentiment Analysis Using BERT Base Supervised Fine Tuning"*, Sensors, 2022

Selected Conference Proceedings

1. Ashim Dahal, Ankit Ghimire, **Saydul Akbar Murad**, Nick Rahimi "*POVQA: Preference-Optimized Video Question Answering with Rationales for Data Efficiency*", arXiv, 2025
2. Ashim Dahal, **Saydul Akbar Murad**, Nick Rahimi "*Embedding Shift Dissection on CLIP: Effects of Augmentations on VLM's Representation Learning*", CVPR, 2025

3. Tomas Nader, **Sadyul Akbar Murad** and Nick Rahimi, "*Human Activity Recognition using an Ensemble Learning Approach*", CATA, 2025 **(Best Paper Award)**
4. Sindhuja Penchala, **Saydul Akbar Murad**, Indranil Roy, Bidyut Gupta, Nick Rahimi, "*Unveiling text mining potential: a comparative analysis of document classification algorithms*", CATA, 2024
5. **Saydul Akbar Murad**, Nick Rahimi *"Secure and Scalable Permissioned Blockchain using LDE-P2P Networks"*, IOTSMS, 2023
6. **Saydul Akbar Murad**, Zafril Rizal M Azmi, Faria Jerin Brishti, Md Saib, Anupam Kumar Bairagi *"Priority Based Fair Scheduling: Enhancing Efficiency in Cloud Job Distribution"*, ICSECS, 2023
7. **Saydul Akbar Murad**, Nick Rahimi, Indranil Roy, Bidyut Gupta *"Design of a New Secured Hierarchical Peer-to-Peer Fog Architecture Based on Linear Diophantine Equation"*, CATA, 2023
8. Apurba Adhikary, **Saydul Akbar Murad**, Md. Shirajum Munir, and Choong Seon Hong *"Edge Assisted Crime Prediction and Evaluation Framework for Machine Learning Algorithms"*, ICOIN, 2022
9. **Saydul Akbar Murad,** Zafril Rizal Bin M Azmi, Nusrat Jahan Prottasha, and Md Kowsher *"Computer-aided system for extending the performance of diabetes analysis and prediction"*, ICSECS-ICOCSIM, 2021
10. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, Md. Al-Imran. *"Comparative Study on Job Scheduling Using Priority Rule and Machine Learning"*, ETCCE, 2021
11. Md. Kowsher, Anik Tahabilder, **Saydul Akbar Murad** *"Impact learning: A Robust machine learning algorithm"*, ICCCM, 2021 **(Best Paper Award)**

Books and Chapters

1. **Saydul Akbar Murad**, Nick Rahimi, Indranil Roy, Bidyut Gupta "*Security challenges in the Industrial Internet of Things (IIoT)*", Quality Assessment and Security in Industrial Internet of Things, 2024
2. Nick Rahimi, **Saydul Akbar Murad**, Sarah B Lee *"Entrepreneurship Opportunities in Cybersecurity"*, Generating Entrepreneurial Ideas With AI, 2024
3. AKM Bahalul Haque, Tasfia Nausheen, Abdullah Al Mahfuj Shaan, **Saydul Akbar Murad** *"Security Attacks and Countermeasures in 5G Enabled Internet of Things"*, 5G and Beyond, 2023
4. Mizanur Rahman, Sajib Debnath, Masud Rana, **Saydul Akbar Murad**, Abu Jafar Md Muzahid, Syed Zahidur Rashid, Abdul Gafur *"Bangla Text Summarization Analysis Using Machine Learning: An Extractive Approach"*, International Human Engineering Symposium, 2023

Job Experience

Jan 2023 – Present	**Graduate Research Assistant** *University of Southern Mississippi, Mississippi, USA.*
May 2021 – Jan 2023	**Graduate Research Assistant** *Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia*
Aug 2022 – Dec 2022	**Lecturer** *International Islamic University Chittagong, Chittagong, Bangladesh*
Feb 2020 – Jan 2021	**Lecturer** *National Model College, Noakhali, Bangladesh*
Feb 2019 – Mar 2019	**Industrial Trainee** *BTCL Nandankanon, Chittagong, Bangladesh*

Teaching Experience

Instructor in the Department of Computer Science at USM

Spring 2026 *CSC 102: Computer Science II*

Spring 2025 *CSC 102: Computer Science II*

Summer 2025 *CSC 327: Secure Software Development*

Fall 2025 *CSC 327: Secure Software Development*

Graduate Teaching Assistant in the Department of Computing Science at USM

Fall 2024 *CSC 428-591: Machine Learning in Cybersecurity*

*2023 – ** *Mentoring undergraduate and graduate students at the Cyber Innovation Lab (CIL).*

Lecturer of Electronic & Telecommunication Engineering at International Islamic University Chittagong

Fall 2022 *ETE–4757: Database Management Systems*

Fall 2022 *ETE-4758: Database Management System Sessional*

Instructor in the Faculty of Computing at Universiti Malaysia Pahang, Pahang, Malaysia

Spring 2022 *BCS-3283: Mobile Application Development*

Lecturer of National Model College, Noakhali, Bangladesh

2020 – 2021 *Data Communication*
Database Management Systems
Programming

Journal Editorship

- Journal of Combinatorial Mathematics and Combinatorial Computing (JCMCC)
  Publisher: Combinatorial Press
- Journal of Computer Science
  Publisher: Science Publication
- Program Committee
  34th SEDE Conference: Software & Data Engineering (SEDE)

Computing Skills

- **ML:** Pytorch, Transformers, Matplotlib, Tensorflow, Keras, Sklearn, NLTK, Pandas, Numpy, Scipy.
- **Simulator:** OMNeT++, CloudSim, iFogSim.
- **Programming language:** C, C++, Java, Python.
- **Database:** MySQL.
- **IDE:** Spyder, Colab, Jupyter Notebook, Code Blocks, Dev-C++, PyCharm.
- **SWE:** Git, Data Structure & Algorithm, HTML, CSS.
- Have strong knowledge on Android app development with Java.
- Have a good Knowledge on P2P communication.

Selected Research & Project

- **EEG to Text Decoding**
  Developed the R1 Translator model, an advanced EEG-to-text decoding framework using LSTM-based encoding and pretrained BART transformer for superior performance. Achieved SOTA results on the ZuCo dataset, surpassing existing models in key translation metrics like ROUGE, BLEU, and CER. GitHub

- **Cyber Threat Detection in Tweets/X**
  Developed a multi-lingual cyber threat detection system for tweets across four languages (English, Chinese, Russian, Arabic) using advanced ML, DL, and LLMs. The research utilized RF, Bi-LSTM, and XLM-RoBERTa models, achieving superior performance in identifying cyber threats. Bi-LSTM outperformed other architectures in terms of classification accuracy and F1-Score. GitHub

- **Preference-Optimized Video Question Answering framework (POVQA)**
  Developed POVQA, a preference-optimized Video Question Answering framework that integrates efficient temporal pooling and rationale supervision to improve data efficiency. By employing advanced pooling techniques and fine-tuning the QWEN-2.5-VL 7B model with SFT and DPO, POVQA significantly enhances performance on the ReasonVQA dataset, achieving F1 scores up to 0.543, BLEU-4 from 0.031 to 0.291, and ROUGE-L from 0.196 to 0.528. GitHub

- **Redemption Score**
  Introduced Redemption Score (RS), a novel hybrid framework for evaluating image captions by triangulating three complementary signals: Mutual Information Divergence (MID), DINO-based perceptual similarity, and LLM-based text embeddings. RS achieves superior human correlation, outperforming traditional metrics on the Flickr8k benchmark with a Kendall-τ of 58.42. This framework provides a more nuanced evaluation by combining visual accuracy and text quality, ensuring robust performance across multiple datasets like MS COCO and Conceptual Captions. GitHub

- **Embedding Shift Dissection on CLIP**
  Analyzed the impact of nine common image augmentation techniques on the representation learning of VLMs, specifically CLIP. The study introduced systematic analysis of embedding shifts, attention map variation, and feature preservation under different augmentations like noise, blur, and perspective transforms. Findings reveal that certain augmentations significantly alter the model's representation, with noise and perspective shift causing the most drastic changes. GitHub

- **ViT vs CNN Segmentation**
  Conducted a heuristic comparison of ViT and CNN for semantic segmentation of remote sensing imagery. Introduced a novel weighted loss function to optimize the Intersection over Union (IoU) and Dice scores while preserving entropy, significantly improving the UNet CNN model's performance. Compared the performance of the UNet CNN with Meta's MaskFormer ViT on the iSAID dataset, demonstrating superior efficiency and inference time in the CNN model. GitHub

- **P2P Communication in Fog Computing**
  This project introduces a new integrated peer-to-peer (P2P) architecture within the fog layer. The proposed model is an advanced iteration of the existing 2-layer LDEPTH architecture. In this sophisticated framework, networks are meticulously organized in a hierarchical manner. GitHub

- **Job Scheduling Technique in Cloud Computing**
  In this project, a novel priority-based job scheduling method, designated as SG-PBFS, has been introduced and meticulously developed. Several experiments have been conducted to validate and refine this technique. GitHub

- **Department Management System Through an Android App**
  This project involved developing an Android application crafted meticulously using Java. The application is designed to serve as a comprehensive educational management tool, encompassing many functionalities. It allows for the efficient addition and management of various academic elements. GitHub

- **Impact learning Implementation**
  Impact Learning is a robust ML algorithm that is widely used to achieve more advanced results on many ML-related challenges. Impact learning is a supervised ML algorithm for resolving classification and regression problems. To install in an ML environment, just command **pip installs impact learning**

Workshops and Presentation

- **Keynote Speaker** of an online academic session on 'Publication and journal selection' of the "Workshop on Research & Publications" at UniV, Bangladesh in 2025.
- **Poster Presentation**, at IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025
- **Keynote Speaker** of an online academic session on 'Publication and journal selection' of the "Workshop on Research & Publications" at UniV, Bangladesh in 2024.
- **Oral Presentation** , at The University of Southern Mississippi, USA on The Topic of Decoding Brainwaves: A Hybrid Neural Network Approach for Translating EEG Signals into Text in 2024.
- **Oral Presentation**, at 2023 10th International Conference on Internet of Things: Systems, Management and Security (IOTSMS).
- **Oral Presentation**, at 8th International Conference on Software Engineering & Computer Systems 2023.
- **Oral Presentation**, at 38th International Conference on Computers and Their Applications, USA.
- **Oral Presentation**, at National Conference for Postgraduate Research (NCON-PGR), 2022, UMP, Malaysia.
- **Oral Presentation**, at 2021 Emerging Technology in Computing, Communication, and Electronics (ETCCE), Dhaka, Bangladesh.
- **Oral Presentation**, at 7th INTERNATIONAL CONFERENCE ON SOFTWARE ENGINEERING & COMPUTER SYSTEM (ICSECS – 2021), UMP, Pahang, Malaysia.

Research Mentorship

May 2024 – Present	**Ashim Dahal, Undergraduate Student** *University of Southern Mississippi, Mississippi, USA.*
Aug 2024 – Present	**Sajim Ahmed, Undergraduate Student** *University of Southern Mississippi, Mississippi, USA.*
May 2025 – Present	**Ankit Ghimire, Undergraduate Student** *University of Southern Mississippi, Mississippi, USA.*
Aug 2025 – Present	**Md Safwan Bin Rashid, Undergraduate Student** *University of Southern Mississippi, Mississippi, USA.*
Jan 2024 – May 2025	**Sindhuja Penchala, Master Student** *University of Southern Mississippi, Mississippi, USA.*
Jan 2023 – Dec 2023	**Israt Jahan, Undergraduate Student** *Noakhali Science and Technology University, Noakhali, Bangladesh.*

Review Research Article

- NeurIPS
- IEEE Internet of Things Journal
- Transactions on Knowledge Discovery from Data
- IEEE Access
- IEEE Transactions on Cognitive and Developmental Systems
- IEEE Reviews in Biomedical Engineering
- Scientific Reports
- ACM Transactions on Knowledge Discovery from Data
- Computer Methods in Biomechanics and Biomedical Engineering
- The Journal of Supercomputing
- Innovations in Systems and Software Engineering
- Journal of Experimental & Theoretical Artificial Intelligence
- Congress on Smart Computing Technologies
- Journal on Artificial Intelligence

Professional Training & Grant Writing

- Graduate Research Ethics Awareness and Training, USM (2025)
- NSF Grant Proposal Writing, USM (2025)
- Grant Writing Workshop, Graduate School, USM (2024)
- High Impact Publication Webinar, UMP (2022)
- Research writing skill, EMK Center (2021)
- Data Science with Python (Part 1), Maker Lab, EMK Center (2020)
- Data Science with Python (Part 2), Maker Lab, EMK Center (2020)
- Android Application Development- Java, SEIP under Basis Institute of Technology and Management (2019)

References

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